

Total No. of Questions : 8]

SEAT No. :

PD4898

[Total No. of Pages : 2

[6404]-455

B.E. (E & TC Engineering)

HONORS IN ROBOTICS

Artificial Intelligence in Robotics

(2019 Pattern) (Semester - VIII) (404183 HR)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data, if necessary.

- Q1)** a) What are the Characteristics of uninformed search algorithms in AI? [6]
b) Explain Hill Climbing Search algorithm? [6]
c) Explain Ant Colony Optimization Technique? [6]

OR

- Q2)** a) What are Characteristics of informed search algorithms in AI? [6]
b) What are Metaheuristics algorithms in AI (list only)? What are its Characteristics? [6]
c) How TABU search optimization algorithm works? [6]

- Q3)** a) What is Concept of Machine Vision? How it is useful in Robotics? [6]
b) How Image processing is useful in Machine Vision and robotics? [6]
c) Explain imaging based automatic sorting and inspection used in Machine Vision? [5]

OR

- Q4)** a) What are Machine Vision algorithms? List out algorithms used in Artificial Intelligence? [6]
b) What are types of Machine Vision techniques for robotics applications? [6]
c) Write a Short note on “Imaging based robot guidance”. [5]

P.T.O.

- Q5)** a) What are the AI tools/techniques used Intelligent robotics systems? [6]
b) What is Task Based Hybrid Closure Grasping Optimization for Autonomous Robot Hand? Explain. [6]
c) Explain accurate Motion Control in Fast Mobile Robots? [5]

OR

- Q6)** a) What are the applications of intelligent systems for mobile Robot Motion Planning? [6]
b) Explain intelligent robotic system for obstacle avoidance? [6]
c) Explain operation of Path Planning Robot Control in Dynamic Environment? [5]

- Q7)** a) How can you correlate AI with Flexible automation? (Any 3 Points). [6]
b) What are advantages of flexible automation? [6]
c) What are applications of FMS? [6]

OR

- Q8)** a) What is process planning in Artificial Intelligence in Flexible Automation systems? [6]
b) What is flexible automation? What are benefits of flexible automation? [6]
c) What is Route optimization for Automated Storage and Retrieval (AS/RS) systems? [6]

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